SHIP BUILDER
HINDUSTAN SHIPYARD LIMITED
VISAKHAPATNAM- 530005.

TENDER TECHNICAL SPECIFICATION FOR PROCUREMENT OF HELO DECK COMMUNICATION SYSTEM (HDCS)

Project	**:**	**DIVING SUPPORT VESSEL VC 11190 – 91**
CLIENT	**:**	**IHQ OF MOD (N)**
DOCUMENT No	**:**	**SOTR/11190-91/005**

000	First Issue to IHQ of MoD (N)		**06.06.2020**	B. Santhi	B. Ravi Kumar
REV	**DESCRIPTION**		**DATE**	**Prepared By:**	**Authorized By:**

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CHAPTER 01 - GENERAL INFORMATION		
Clause No.	Description of Technical Specification	Vendor Reply Complied / Not Complied
1.1	**Introduction:** The Helicopter Deck Communication is intended for use on the flight deck and compartments of helicopter carrying ships as per the system configuration defined herein. Portable sets would be used for communications on the helicopter deck by all members of the Flight Deck Team and fixed equipment will be used in closed compartments of the ship. The equipment shall be capable of hands free operation on the flight deck. The set shall be water resistant /marine conditions proof and would have been tested for operation in a harsh sea environment (salt spray and high velocity winds).	
1.2	**Aim:** The aim of this SOTR is to lay down the user requirements and technical parameters for a Helo Deck Communication System (HDCS) based on contemporary TETRA technology capable of both fixed and hands-free communication. Further, this would be an independent system for communications required for safe and efficient, conduct of flying operations.	

CHAPTER 02 - OPERATIONAL REQUIREMENTS		
Clause No.	**Description of Technical Specification**	**Vendor Reply Complied / Not Complied**
2.1	**System Characteristics.**	
2.1.1	The complete system should be COTS based, modular and based on TETRA technology.	
2.1.2	The fixed sets should have fixed microphone and speakers with facility to plug in headsets.	
2.2	**System Configuration.**	
2.2.1	The indicative list of positions that are required to be connected in the HDCS are appended below. However, the exact number of fixed and portable positions would be defined and decided separately depending on type and role of the ship. There is a requirement for communications between individual positions as well as from one position to multiple positions:-	

S.No.	Position	Remarks
(i)	LSO Cabin	01 fixed
(ii)	Bridge	01 fixed
(iii)	HELO deck	Chocks & lashing Party – 04 (Portable). HELO deck F/F including CFD – 10 (Portable)
(iv)	OPS Room	01 fixed
(v)	Air Crew Room hands free	01 Portable
(vi)	AVCAT Pump Room	01 fixed
(vii)	Spare-hands free	01

2.2.2	The typical hierarchy of positions in the HDCS would be as follows:	
(a)	Priority 1 – LSO Responsible for control of flying operations	

	and deck operations.	
(b)	Priority II-CFD Responsible for HELO deck Operations.	
(c)	**Priority III - Bridge and Ops Room** Responsible for coordination between Hangar Control and CFO for flying operations and coordination with other departments.	
2.2.3	The actual hierarchy of positions in a ship is to be configurable, default hierarchy positions will be frozen at the time of order placement.	
2.2.4	The rest of the personnel manning the HDCS are to be put on the lowermost rung of the hierarchy, Priority for interrupting communication is to be given to personnel placed higher on the ladder of hierarchy.	
2.2.5	Suitable method for switching between channels is to be provided.	
2.3	**Portable Units (Hands Free Equipment - Headset).** The requirements are as follows:-	
2.3.1	The headset for portable wireless unit should be helmet mounted and should be cordless, except cord or wire between clip-on module (secured on waist) and the microphone-speaker.	
2.3.2	The length of cord as mentioned above must cater to a maximum user height of 200cm.	
2.3.3	Noise suppression filter/mechanism provided should be able to suppress the ambient noise level of about 150 dB to a safe limit of 80 dB or better.	
2.3.4	Should have a minimum operating range of 50 meters, thereby covering the extremities of the Helo deck and ACR (Air Crew Room).	
2.3.5	User should be able to switch ON/OFF, control volume and select channel on the mobile units.	
2.3.6	Portable wireless units should be compact, lightweight (<1Kg) and powered by rechargeable batteries with following usable time :- (i) Standby Time of 12 hours or better	

	(ii) Talk Mode of 2 hours or better	
2.3.7	The portable wireless units should have the necessary securing arrangements to prevent risk of Foreign Object Damage (FOD) such as clips or metallic securing chain. This will include arrangement for securing the portable units to the user through strap(s) and to designated stowage position onboard ship.	
2.3.8	Knobs/Push buttons should be designed for easy operation and identification even in pitch dark conditions.	
2.4	**Additional mobile units and Batteries.** At least 2 sets of additional batteries should be provided for each mobile user. At least two Battery charging stations with provision to simultaneously charge batteries for upto all 16 mobile users should be supplied as part of equipment. It is to be ensured that batteries do not emit spark during the battery charging operation to prevent accidental ignition. The system should be Intrinsically Safe (IS) TETRA terminal approved by any international accredited certifying agency.	
2.5	**Fixed Units.** Fixed Units should have in-built speakers, microphone and pluggable headsets. Noise suppression filter / mechanism provided should be able to suppress the ambient noise of about 150 dB to safe limit of 80 dB or batter.	
2.6	**Warning Indications.** Suitable audio-visual warning should be provided on the mobile units for low battery indication.	
2.7	**Recording Facility.** The system should have dedicated recording facility with following provisions: - (a) Recording and replay facility of all channels for at least one Terabyte with time tags. Recording should be transferable to secondary storage device like Compact discs, DVD etc. (b) The recording should be in standard contemporary audio formats compatible with commercially available audio players and associated software.	
2.8	**Lifting Arrangement** Units weighing more than 40 k.g, if any, shall be provided with collar eyebolts or suitable lifting lugs / arrangements. If the eyebolts cannot remain in situ after the unit	

	has been installed on the ship, provision is to be made for securing them on the unit.	
2.9	**Terminal** Bolted type terminal and crimped socket of electrolytic copper are to be provided for all incoming and outgoing cables. All cables to be marked with Aluminum tallies indicating the route of cable.	
2.10	**Anti-Condensation Heating.** Wherever necessary, equipment shall be provided with anti-condensation heating elements, which are to be switched on automatically when the equipment is switched off. Manual override for automatic switching ON of Anti Condensation heating elements, when the equipment is switched off, is to be provided.	
2.11	**Safety Standards.** The system should offer total safety to personnel from shock and comply to IEC/ETSI EN 60950 standard or equivalent. All fixed units are to be provided with earthing bolt. Where units are connected with the ship's main, fuse protection is to be provided. Each unit where dangerous voltages are present shall bear a Red Label stating in bold white characters the highest voltages inside the unit. Doors and panels of equipment enclosures are to be provided with safety door switches, wherever hazardous voltages are present inside.	
2.12	**Cabinet Cooling.** Adequate precautions need to be taken during the design of the cabinets to ensure sufficient cooling and circulation of air. Additional provisions for cooling are to be provided to ensure optimal system performance and to prevent defects caused due lo overheating under ambient conditions as mentioned in the environmental specifications.	
2.13	**Shock Mounts.** The fixed equipment would be mounted on shock mounts onboard the ship. Suitable shock mounts are to be provided by the firm.	
2.14	**Test Points.** All test points, indicators and controls shall be suitably labelled and preferably provided on the front panel for easy access and the same to be elaborated in the concerned document. All connectors required for testing need to be supplied with the equipment.	
2.15	**Knobs/Push Buttons.** Should be designed for easy operation	

	and identification even in pitch dark conditions.	
2.16	**<u>Antenna System</u>.** Antenna system, if and as required, is to be provided for meeting system operation requirements as mentioned above.	
	Maintenance:	
2.17	**<u>Test equipment</u>.** The manufacturer would have to provide details of test equipment required for onboard maintenance.	
2.18	**<u>BITE.</u>** The system should have the capability for on line and offline maintenance diagnostics covering all modules and PCBs. The facility for conclusive checks of the health monitoring system itself should also be available. The facility of audio-visual alarms in case of failure of any modules and PCBs is also essential.	
2.19	**<u>Degraded mode.</u>** The provision of operating the system in degraded mode and graceful degradation of performance should also be incorporated in the design to ensure that defect/failure of one module/PCB does not cause the non-availability of the complete system.	

CHAPTER 03 - TECHNICAL SPECIFICATIONS

Clause No.	Description of Technical Specification	Vendor Reply Complied / Not Complied
3.1	**Technology**. The system should be based on TETRA technology.	
3.2	**Frequency Range**. V/UHF 400-430 MHz	
3.3	**Frequency Stability**. ≤ 1ppm	
3.4	**Channels**. At least 3 channels catering for 16 wireless positions.	
3.5	**RF Power Output** Less than 5 W.	
3.6	**Security**. The system should provide E2E, AES 256 encryption or better.	
3.7	**Power supply**. The system should be designed to work on 230V, 1 Ph, 50Hz power supply of the ship. The system should have provision for operating on 24V DC supply in case of emergency. The equipment should be able to withstand fluctuations of 10% in voltage and 3% in frequency. In case system needs any other power supply, the same needs to be catered for, wherever required, through the use of suitable converters/transformers etc., which are to be included in the scope of deliverables.	
3.7.1	Distributed UPS complying with EED 50 -77 for back up supply for the equipment be included in the SOTRs iaw IHQ (N)/DEE policy letter EE/03/9711/Policy/L-105 dated 24 Jan 17.	
3.8	**Headset and Helmet.** The headsets and helmet should be compatible with the portable wireless units and will be part of the scope of supply. The helmet should have following features:-	
3.8.1	It should be impact resistant and should provide protection against head/skull injury and burns due to flash. (cranial type helodeck safety helmets)	
3.8.2	It should be light weight and comfortable to wear for prolonged duration,	
3.8.3	It should provide Electrical shock protection.	
3.8.4	Integrated sun glare wind dust and FOD protection visors to be provided for both day and night operations.	
3.8.5	It should be compatible with wireless based communication equipment with a noise cancelling microphone/laranga phone (with both voice activated and PTT facility).	
3.8.6	Provide double hearing protection.	
3.8.7	Ear cup cushion and foam to be of lasting quality with	

	minimum life of 2 years under marine usage.	
3.8.8	The system cables as required for interconnection of various sub-units of HDCS is under Supplier’s scope. The length of system cables would be estimated by HSL based on the layout and cable routing during detail design and forwarded to the supplier for supply of cables accordingly. Per meter cost of each type of system cables shall be indicated in the price-bid. HSL shall provide the power supply cables only for the system.	
3.8.9	All power cables should comply with EED 50-12 (Rev-2) and internal cabling should be carried out using LFH Cables.	

<table>
<tr><th colspan="3">CHAPTER 4 – ENVIRONMENTAL SPECIFICATIONS</th></tr>
<tr><th>Clause No.</th><th>Description of Technical Specification</th><th>Vendor Reply Complied / Not Complied</th></tr>
<tr><td>4.1</td><td><u>Type Test</u> The system is to be subjected to type testing as per the specifications given below:-</td><td></td></tr>
<tr><td>4.2</td><td>
<table>
<tr><td>Sl. No.</td><td>Test</td><td>Specification</td></tr>
<tr><td>(a)</td><td>Seaway Condition Test</td><td>(i) Roll – max ±30º with 8 sec period-Operational
(ii) Pitch – max ±10º with 20 sec period-Operational
(iii) List –max 20º from vertical (permanent) – survival
(iv) Trim – max 5º</td></tr>
<tr><td>(b)</td><td>Damp Heat</td><td rowspan="4">IEC 60068 – 2.2 / equivalent (Severities applicable for Merchant Ships)</td></tr>
<tr><td>(c)</td><td>Dry Heat</td></tr>
<tr><td>(d)</td><td>Cold</td></tr>
<tr><td>(e)</td><td>Salt Mist</td></tr>
<tr><td>(f)</td><td>Humidity and Heat</td><td>IEC 61347-1 Clause 1.1 or equivalent (Severities applicable for Merchant Ships)</td></tr>
<tr><td>(g)</td><td>Ingress Protection</td><td>All units installed on the weather –deck of intended for use on the weather –deck should be IP57 compliant and for use in below deck compartments should be IP54 complaint. Test as per IEC 60529/equivalent.</td></tr>
<tr><td>(h)</td><td>Shock</td><td>IEC 68227 or equivalent</td></tr>
<tr><td>(J)</td><td>EMI/EMC</td><td>IEC 60945 or equivalent</td></tr>
</table>
</td><td></td></tr>
</table>

CHAPTER 5 – RELIABILITY AND MAINTAINABILITY SPECIFICATION		
Clause No.	**Description of Technical Specification**	**Vendor Reply Complied / Not Complied**
5.1	**Reliability and Maintainability.** The system should have reliability greater than 95% and MTBF of at least 10,000 hours. The MTTR should not exceed 20 minutes.	
5.2	**Product Support.** The supplier shall confirm the product support for a minimum period of 10 years for the equipment offered by them from the date of commissioning of the system. Essential infrastructure established ashore will be such that it can sustain 10 years from the time of commissioning of the last equipment. The manufacturer/ Production Agency will provide life time product support including: (a) Supply of spares (b) Up-gradation when available (c) Modification and repairs. (d) Three (03) years notice, in case of any likely production shut down, to enable procurement of L TE spares. (e) Undertaking repairs (Beyond Local Repairs), after the warranty period and continued product support for 10 years after the supply of equipment.	

CHAPTER 6 – GENERAL REQUIREMENTS		
Clause No.	**Description of Technical Specification**	**Vendor Reply Complied / Not Complied**
6.1	**Supplier Scope of Supply, Installation and Commissioning:** The supplier's scope of system supply shall cover the supply of equipment, Installation Material, Spares, Drawings and Documentation, installation supervision, setting to work, testing, on board commissioning, harbor and sea trials of HDCS. The supplier is to list down in detail, the deliverables to be provided to the customer at various stages of supply as mutually agreed upon. The complete scope of supply of the system by supplier shall include as given in succeeding paragraphs.	
6.2	**<u>Quantity:</u>** One (01) HDCS per Ship. Total Requirement: Two (02) HDCS for 02 Ships.	
6.3	**Installation Material:** One set of installation material that includes inter cabinet cables, connectors, shock mounts, special fasteners, special tools, special fittings etc. shall be supplied by the supplier.	
6.4	**Testing and Tuning Spares:** The supplier shall recommend one set of Testing and Tuning Spares required for STW, HATS and commissioning of the system.	
6.5	**Onboard Spares:** OEM recommended onboard spares for schedule servicing, maintenance & break down maintenance falling due within the first operation cycle for 02 years shall be provided for the equipment in ILMS Ver 2 format (Integrated Logistics Management System) indicating the item code number, quantity and item wise cost. The Spares consumed within the guarantee period to rectify / liquidated known defects shall be replaced free of cost. The supplier shall prepare the list of OBS (including onboard maintenance tools) based on the Reliability and Maintainability data and taking into account IN's maintenance philosophy, and shall forward to IHQ MOD (Navy) for vetting. The complete inventory of system parts including OBS & B&D spares is to be provided in INCAT (Indian Naval Catalogue of Inventory) compatible format in electronic media for ILMS (Integrated Logistics Management system) of Indian Navy, for the management of spares. The supplier shall provide both hard copy as well as soft copy. Refer Annexure-1	

6.6	**Five Year Base and Depot (B&D) Spares.** Base spares recommendation shall cover maintenance / overhaul requirements of 5 years including two refits shall be provided for the equipment in ILMS Ver 2 format. Recommendations for insurance holding of long lead items are also to be indicated. Itemized cost of B&D spares are to be forwarded with the main offer along with part identification list of spares. The list would contain price, description, Pattern no. and quantity fitted on each equipment in respect of various parts / components. The quotation of spares shall be valid for at least 18 months. The Buyer shall subsequently range and scale the B&D spares. The cost negotiation of B&D spares would be carried out by a committee constituted by the Buyer, wherein the seller's and sub-vendor's representative would be a member. The OEM shall undertake procurement of the ranged and scaled spares as per the ordering instructions issued by the IHQ MoD (N). Refer Annexure-1	
6.7	**Documentation:** Separate manuals in respect of following for both equipment and system shall be provided in hard and IETM level 4 format:- Refer Annexure-2 (a) Operation and Specifications including onboard Repair and Maintenance. This should include information about the equipment (such as serial number, trial reports, etc) and the manufacturer (such as addresses, telephone and fax numbers, email, etc). (b) Repair Technical Document with detailed drawings (c) Complete Parts List	
6.8	**Binding Data:** Three hard copies and two sets on CD ROM of the following binding drawings / documents are to be supplied by the OEM within 01 week of placement of order:- (a) Block diagram of the system (b) Installation documents covering detailed procedure for installation with sequence of activities. (c) Installation drawings indicating overall dimensions, CG, weight, maintenance envelope etc of each unit. (d) Recommended arrangement of devices in nominated compartments. (e) Inter unit-cabling diagram with cables specification. (f) Cable connection schedule (g) Power supply scheme for the system (h) Heat dissipation of individual units in compartment &	

	in close loop ventilation system, as required for the system. (i) Parts identification list indicating part no, qty, maker's name, specification etc. (j) Requirements for support systems such as fire fighting, communication, lighting etc. (k) Cable length limitation etc, if any. (l) Detailed foundation drawing including bolting plan.	
6.9	**Installation:** Onboard installation of the system will be carried out by the Shipbuilder. The assistance and responsibilities of the OEM include the following:- (a) Coordination with the Shipbuilder for the installation work. (b) Providing instructions to the Shipbuilder personnel for installation of the system. Cable laying will be done by the shipbuilder before start of the installation activities. (c) Checking of the cable connections (cold wire test) Providing details of cable run from Antenna to below deck compartments.	
6.10	**Setting –to-work:** The supplier shall carry out:- (a) Setting-to-work. **(b)** Testing and Tuning of the system.	
6.11	**Service Engineer:** Vendor shall indicate lumpsum charges for connectorization, STW, HATs and SATs at yard. Vendor shall forward the major check points where the service engineer is required along with the no. of days at each stage. The check list of jobs to be completed by HSL prior to arrival of service engineer for connectorization and commissioning is to be submitted within Two (02) weeks of placement of order. **Note:** In case the check list is completed by HSL and the work is not completed within envisaged period no extra payment would be made by HSL. However, for delay attributable to HSL, extra fixed manday rate is to be submitted along with the bid.	
6.11.1	The complete system of the equipment shall be installed on board the vessel under the complete responsibility of the suppliers and for this purpose, suppliers shall depute their service engineer to supervise the installation work, along with necessary service manuals, spares, tools etc.	
6.11.2	The cable laying shall be carried out by HSL and the connectorization & connectors, STW, HATs and SATs are in OEM Scope.	

6.11.3	**FATs and HATs/SATs.** The supplier is to offer the system for FATs, HATs and SATs as per IHQ, MoD (N) approved schedule.	
6.12	**FATs/IIs/HATs/SATs**. The complete system would be subjected to Factory Acceptance Trials (FATs)/Installation Inspections (IIs) / Harbour Acceptance Trials (HATs)/Sea Acceptance Trials (SATs), as applicable by IN norms, in India. The basic checks that will be offered during the trials should be included in the technical proposal. The detailed methodology and schedule for each of these trials is to be formulated by the manufacturer and provided to IHQ MoD (Navy) at least 6 months prior to the commencement of each of these trial phases, so as to be amended / approved by IHQ MOD (Navy) and then formalized as the acceptance schedules by IN, for conducting FATs/IIs/HATs/SATs.	
6.13	All EMI/EMC and Environmental Qualification tests as per previous chapter respectively need to be completed prior to FATs.	
6.14	**Harbour Acceptance Trails (HATs).** The supplier is required to provide necessary representative(s) to assist during the following phases:- HATs/SATs protocol shall be furnished for IHQ MOD (N) approval.	
6.15	**System Performance Responsibility.** In case of any irregularities in the operation/performance of the system or non-conformance to specified parameters observed on integration with ships system, the supplier is bound to rectify the defect. The supplier shall ensure complete responsibility of satisfactory operation of the system on board.	
6.16	**Maintenance**. The manufacturer should provide recommended maintenance schedules for preventive and corrective maintenance of the system.	
6.17	**Training:** Mandatory training shall be imparted to the ship's crew and maintainers for the operation and maintenance of electrical equipment installed on board. OEM shall arrange for such training to ship's crew and maintainers prior to delivery of the DSV. The crew are required to be trained for equipment operation and on board repairs while the maintenance personnel are to be trained for planned preventive maintenance and other repairs of the equipment. The training shall be arranged at OEM premises and onboard/HSL premises. The details of training required on equipment shall be finalized during the TNC / ordering of the equipment.	

6.18	**Guarantee:** As per Commercial terms and conditions.	
6.19	**Shipping**. All equipment shall be adequately packed and protected with supports to ensure adequate protection during all modes of transportation. Each unit within a package / container shall be clearly marked for identification. The container shall clearly indicate the item description with caution marks, quantity, weight, size etc. A separate document giving details and instructions for storage, preservations, handling and transportation after delivery is to be supplied. The supplier should indicate the delivery schedule, transport, packing, preservation, insurance etc.	
6.20	**Upgrading of System**. The manufacturer should give an undertaking in the contract, to make available all future upgrades to the system software and hardware. As far as possible, such upgrades should be possible, with minimal change of system configuration. The OEM should also undertake to undertake upgradation of the system and modifications to the system software should it be necessary to interface additional/alternate systems at a later date.	
6.21	**Preservation, packing and Shipping**: The stores (including OBS and B&D Spares) shall be supplied in long-term preserved condition that is suitable for storage under tropical high humidity conditions for a period of 24 months.	

FORMAT FOR MANUFACTURE'S RECOMMENDED LIST OF SPARES,TOOLS, SOFTWARE(MRL-OBS) AND BASE & DEPOT SPARES(B&D)- (ILMS FORMAT) : **- Annexure - 1**

	Vendor Master	Vendor Master	Vendor Master	Vendor Master	Vendor Master	Vendor Master	Material Master	Material Master	Material Master	Material Master	Material Master	Material Master	Material Master	Material Master	Material Master	Material Master	Material Master	Material Master	Material Master	Material Master	Material Master
Col No..	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Data source..	OEM	OEM	OEM	OEM	OEM	Navy	OEM	OEM (Equip)	OEM	OEM	OEM	Navy	OEM/Navy	OEM/ Navy	Navy	Navy	Navy	OEM	OEM	OEM	OEM
Column title…	OEM Name	OEM Address	OEM Telephone No	OEM Fax	OEM Email Addresses	Navy Vendor Code	Vendor em Code	Equipment Item Code	Item Code	Item Desc	Item Demo	Active/Inactive	CRP category	VED Category	Material Group	Material Type	Station Code	Price	Currency	Price Reference	Price date
Column Description	OEM Name	OEM Address	OEM Telephone No	OEM Fax	OEM Email Addresses	Vendor Code assigned by the Indian Navy	Complete item Code assigned by the OEM for Eqpt. Assy , Sub-assy. Kit spars s etc	Eqpt identification no.assigned by OEM for Main Equipment only	Item identification no. assigned for Eqpt. Assy. Sub assy. Kit spares parts etc by OEM truncated to Max.32 Char Item code should only contain' alphanumeric characters 'and 'tryphen'(-),itemcode, Vendor ccc coo Char'08'and itemcode from 'g' to '40'max	Description of items/Eqpt/Assy/Sub Assy/Stockable /Kit.	Unit of measurement for the items	Type active or inactive 1.Active:Item present in ILMS System. 2.inactive:Item not present in ILMS System.	Nature of item Consumable© /Returnable®/Permanent(P).To be recommended by OEM as per defintion in explanation sheet and approved by Navy.	Vital(V)/Essential€/ Desirable(D) Recommended by OEM as per definition in explanation sheet and approved by Navy.	Section Head	Type of Material	Station in which material is present	Price	Currency for Price	Price Reference (Order details /OEM Price List Details)	Date as per price Reference
Data Structure	CHAR(255)	CHAR(255)	CHAR(255)	CHAR(255)	CHAR(255)	CHAR(6)	CHAR(255)	CHAR(40)	CHAR(40)	CHAR(255)	CHAR(3)	CHAR(1)	CHAR(1)	CHAR(1)	CHAR(10)	CHAR(10)	CHAR(4)	NUM(18,0)	CHAR(5)	CHAR(60)	CHAR(15)

[cont………]

							Equipment BQM	Equipment BQM	Equipment BQM	Equipment BQM	Equipment BQM	Equipment BQM	Equipment BQM	Equipment BQM	Equipment BQM	Equipment BQM	Equipment BQM
Material Master	Material Master	Material Master	Material Master	Material Master	Material Master	Material Master	Material BQM	Material BQM	Material BQM	Material BQM	Material BQM	Material BQM	Material BQM				
22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39
OEM	OEM	OEM	OEM	OEM	OEM	OEM	OEM	OEM	OEM	OEM	OEM	OEM	OEM	OEM	OEM	OEM	OEM
Manufacturer Name	Auth Supplier Name	Specification	Characterstics	Drawing Reference	PIL Reference	Unit Weight	EQuipment	A-Assembly	S-SubAssembly	L-Stockable	KIT	Superior Element	PIL Qty	OBSQty	B&Dqty	LTE qty	MRLSQty (Ranging &scaling)
Manufacturer Details (Name) for item other than an Eqpt (in case different from OEM of Main Eqpt)	Supplier Details (Name of auth supplier) for Eqpt/assy Sub-assy/Stockable/Kit in case different from OEM	Specifiaction (Any standard Spec or Material Spec which)f or the item	Contaned the properties/characteristics of the corresponding item viz dimension etc.	Drawing Reference	PIL Reference (PIL No. Version Page and Line no.)ht	Unit Weig.	If item Code is a main Equipment then type Y,else leave the ce;; blank.	1. If item is an Assembly,then type Y else N 2.If value is 'Y' then fill up column Named "Superior Element with corresponding Equipment Code.	1.If item Code is an SubAssembly then typr Y,else type N. 2.If value is 'Y' then fill up column Named 'Superior Element with corresponding Assembly identification no.orthe Equipment identification no.in casxe item is not part of any Assembly.	1.If item Code is a Spare partthen type Y,else type N. 2.If value is 'Y' then fill up clumn Named 'Superior Element with corresponding Sub-Assembly identification no.or Assembly identification no.of Assembly identification no.or the Equipment Code (in case item is not part of any sub-assembly or Assembly).	1.If itemCode is a KIT,then type Y,else leave the cell blank. 2.If value is Y,then fill up column named 'Superior Element with corresponding Equipment Code. 3.List of Constituent part no. and Qty comprising the kiy is to be forwarded as separate sheet for each list Refer Annexure 'A'	Indicates the higher linked Sub-Assembly or Assembly or Assembly or Main Equipmento which the itemcode belongs	Enter the PIL quantity of material which is mentioned column item Code,Quantity of item fitted in particular equipment	Enter the OBS quantity of material which is mentioned in column item Code	Enter the B&D quantity of material which is mentioned in column item Code	Enter the LTE quantity of material which is mentioned in column item Code	Qty indicated in Manufacturer recommended List of spares for the item in column item Code
CHAR(2	CHAR(	CHAR(	CHAR(2	CHAR	CHA	CH	CHAR(1	CHAR(1	CHAR(1)	CHAR(1)	CHAR(1)	CHAR(40)	NUM(18,0)	NUM(18,0)	NUM(18,0)	NUM(18,0)	NUM(18,

55)	255)	255)	55)	(255)	R(25 5)	AR(8)	)	)									0)
Note: All data are entered just for examination.																	

VED – VITAL/ESSENTIAL/DESIRABLE analysis of spares to be carried out by OEM prior to submission to the BUYER.

Notes: -

_Data regarding maintenance spares/stores like lubricants, sealing compound, gases should be given separately giving source of supply.

Data furnished as OBS and B&D should also include software backups, as applicable.

In ‘Remarks’ column following information (if applicable) be given: -

If an item has a shelf/operational life it be marked as ‘G’ and life indicated.

Matching set of components be indicated.

Item which can be locally manufactured should be marked ‘LM’

Items which cannot be manufactured in India due to sophisticated design/technology may be marked as ‘SI’ (Special item).

If a component/assembly is common to other similar equipment offered by the OEM earlier, these should be marked ‘CM’ and name of the equipment be indicated.

OBS and B&D spares list should be drawn out of the ‘Part list’ of the equipment, which should be separately given as part of Technical Manuals.

If the main equipment consists of other, then OBS and B&D spares list should be prepared for them under proper heads.OBS and B&D spares list is to be per the maintenance concept of the customer.

Items provided along with the equipment as spares should be included in OBS and B&D list.

Modules/Shop Replaceable Unit (SRU)/ assemblies should be listed and their components should be listed components should be included under them so as to relate each item of spare to their module/SRU/ assembly.

OBS and B&D list for test equipment should also be provided on the similar format.

Buyer would also have the option to amend the MRL-OBS proposed by the bidder during the Technical Negotiation of individual equipment to ensure its sufficiency, based on its past experience of exploitation of same/similar equipment.

Cost to be indicated in Price bid only.

Sufficiency clause: - the seller would either ‘Buy Back ‘the spares rendered surplus or exchanges them on cost to cost basis with the spares as required by the Buyer. The said spares would be purchased or replaced by the seller, based on the prices negotiated in the contract.

Annexure-2

TECHNICAL DOCUMENTATION
(To be filled by OEM)

Name of Vessel/Equipment: ________________
Original Equipment Manufacturer (OEM): __________________

S.No	Technical Literature	Unit Cost	Scale For 02 Vessels	Total Cost	Remarks
1.	User Handbook/Operator's Manual				
2.	Design Specifications				
3.	**Technical Manual** **(a) Part I** Tech description, specifications, functioning of various Systems. **(b) Part II** Inspection /Maintenance tasks Repair procedures, materials Tools (SMT's)/ Special Test Eqpt (STE)s **(c) Part III** Procedure assembly/disassembly, repair up to component level, safety precautions. **(d) Part IV** (i) Part list with drawing reference (ii) List of SMT/STEs with Test Bench				
4.	Manufacturer's Recommended List of Spares (MRLS)				
5.	IIlustrated Spare Part list(ISPL)				
6.	Technical Manual on STE with drawing reference.				
7.	Soft copy on the above Tech literature				
8.	Any other (specify)e.g. Service logs etc.				

Total Cost

Notes: - 1. In case any additional equipment is used, their tech literature will be included.
2. If certain technical literature is being provided free of cost, it should be indicated in the remarks column.
3. Cost to be indicated in Price Bid Only.

DISTRIBUTION OF DOCUMENTS

S.No	DISTRIBUTION OF DOCUMENTS	TYPE OF DOCUMENTS	No. OF COPIES	Vendor COMPLIANCE (Y/N)
1.	FAT PROTOCOLS FOR OWNERS APPROVAL	CD	2	
		PRINT	5	
2.	GRAQ PROTOCOLS FOR OWNERS APPROVAL	CD	2	
		PRINT	5	
3.	SCHEMATIC SYSTEM DRAWINGS	CD	2	
		PRINT	9	
4.	AS MADE EQPT. DRAWINGS	CD	2	
		PRINT	10	
5.	CPL/PIL	PRINT	8	
6.	EQPT HAND BOOK TECH.OPERATION MANUAL	CD	2	
		PRINT	10	
7.	INSTALLATION SPECIFICATION MANUAL	CD	2	
		PRINT	10	
8.	EQPT.FIT BOOK	CD	1	
		PRINT	7	
9.	TEST CERTIFCIATES	CD	3	
		PRINT	7	
10.	TRIAL REPORTS-FAT,HAT&SAT	CD	2	
		PRINT	7	
11.	MATERIAL SCHEDULE	PRINT	6	
12.	MAINTENANCE SCHEDULE	PRINT	10	